The empirical recurrence for OEIS A297821, recovered from its tail

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Abstract

OEIS A297821 records "Empirical recurrence of order 98" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 231$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 10$, so the recurrence is proved for every $n \geq 108$. Its last coefficient is $c_{98} = 4 \neq 0$, so no further tail satisfies anything shorter; any order-98 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A297821 is "Number of nX6 0..1 arrays with every element equal to 1, 2 or 3 king-move adjacent elements, with upper left element zero.". It has offset 1 and begins

$$5, 573, 1114, 4306, 14237, 46701, 146530, 456814, \ \ldots$$

The entry states:

> Empirical recurrence of order 98 (see link above)

As of the "Last modified" line on the live entry (Jan 06 2018 10:57:45, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 231$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 231$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 98: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 10$ is the first whose minimal order is exactly the stated 98.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 462$ exact terms from $n = 10$ on, it returns order 98 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{98} c_i\, a(n-i),$$

c_{1}	10	c_{26}	−45769	c_{51}	−324629	c_{76}	35051
c_{2}	−41	c_{27}	69258	c_{52}	−49355	c_{77}	−157523
c_{3}	92	c_{28}	17763	c_{53}	312925	c_{78}	38693
c_{4}	−116	c_{29}	−262802	c_{54}	738059	c_{79}	12405
c_{5}	27	c_{30}	−915	c_{55}	−1361822	c_{80}	57132
c_{6}	223	c_{31}	65161	c_{56}	95281	c_{81}	−47607
c_{7}	−661	c_{32}	−49640	c_{57}	824724	c_{82}	−23522
c_{8}	1391	c_{33}	−414297	c_{58}	1778049	c_{83}	−8948
c_{9}	−2365	c_{34}	448392	c_{59}	−742156	c_{84}	6972
c_{10}	3315	c_{35}	−135576	c_{60}	−849765	c_{85}	8504
c_{11}	−3279	c_{36}	709977	c_{61}	189645	c_{86}	2360
c_{12}	2051	c_{37}	−588442	c_{62}	−931200	c_{87}	3440
c_{13}	−27	c_{38}	1233776	c_{63}	3987	c_{88}	−90
c_{14}	−1502	c_{39}	−188157	c_{64}	−1093028	c_{89}	865
c_{15}	1921	c_{40}	474014	c_{65}	1544364	c_{90}	−464
c_{16}	−7620	c_{41}	−1111097	c_{66}	−482991	c_{91}	−469
c_{17}	16110	c_{42}	550138	c_{67}	897265	c_{92}	−386
c_{18}	−27739	c_{43}	−881363	c_{68}	−962342	c_{93}	−153
c_{19}	15602	c_{44}	−777473	c_{69}	471549	c_{94}	−42
c_{20}	−5187	c_{45}	−102723	c_{70}	−383474	c_{95}	39
c_{21}	44956	c_{46}	85955	c_{71}	−680	c_{96}	16
c_{22}	−84575	c_{47}	317307	c_{72}	180284	c_{97}	10
c_{23}	104924	c_{48}	−858708	c_{73}	62756	c_{98}	4
c_{24}	10408	c_{49}	744482	c_{74}	318659		
c_{25}	−28354	c_{50}	703453	c_{75}	−204669		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 108$, and it is the recurrence OEIS A297821 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 10$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{98} - c_1 x^{98-1} - \cdots - c_{98}$. Its constant term is $-c_{98} = -4$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 98 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 98, so $\deg q = 98$, and it is monic. It holds from some index t on. From $\max(t, 10)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 98, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 22 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 98, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 4.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-98 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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